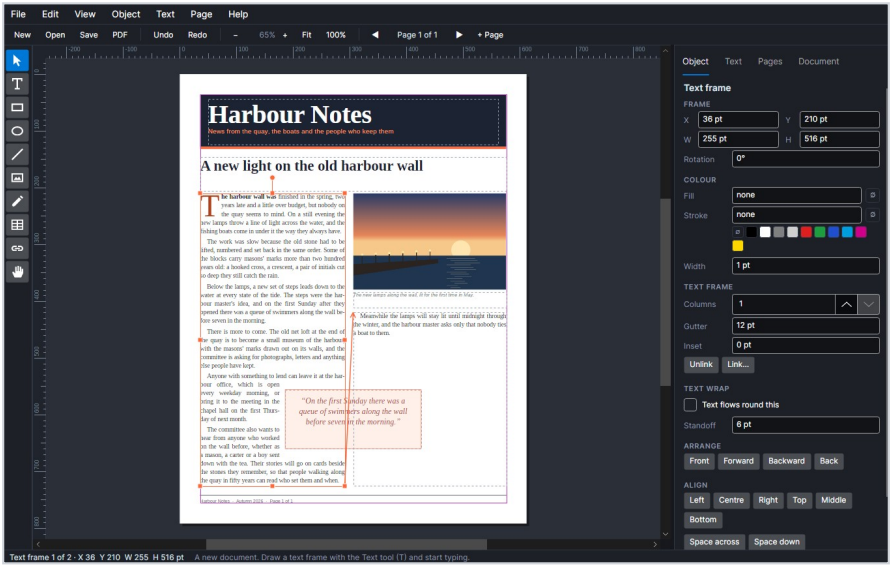
AVAPageStream

User Guide

A desktop publishing program



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Getting started

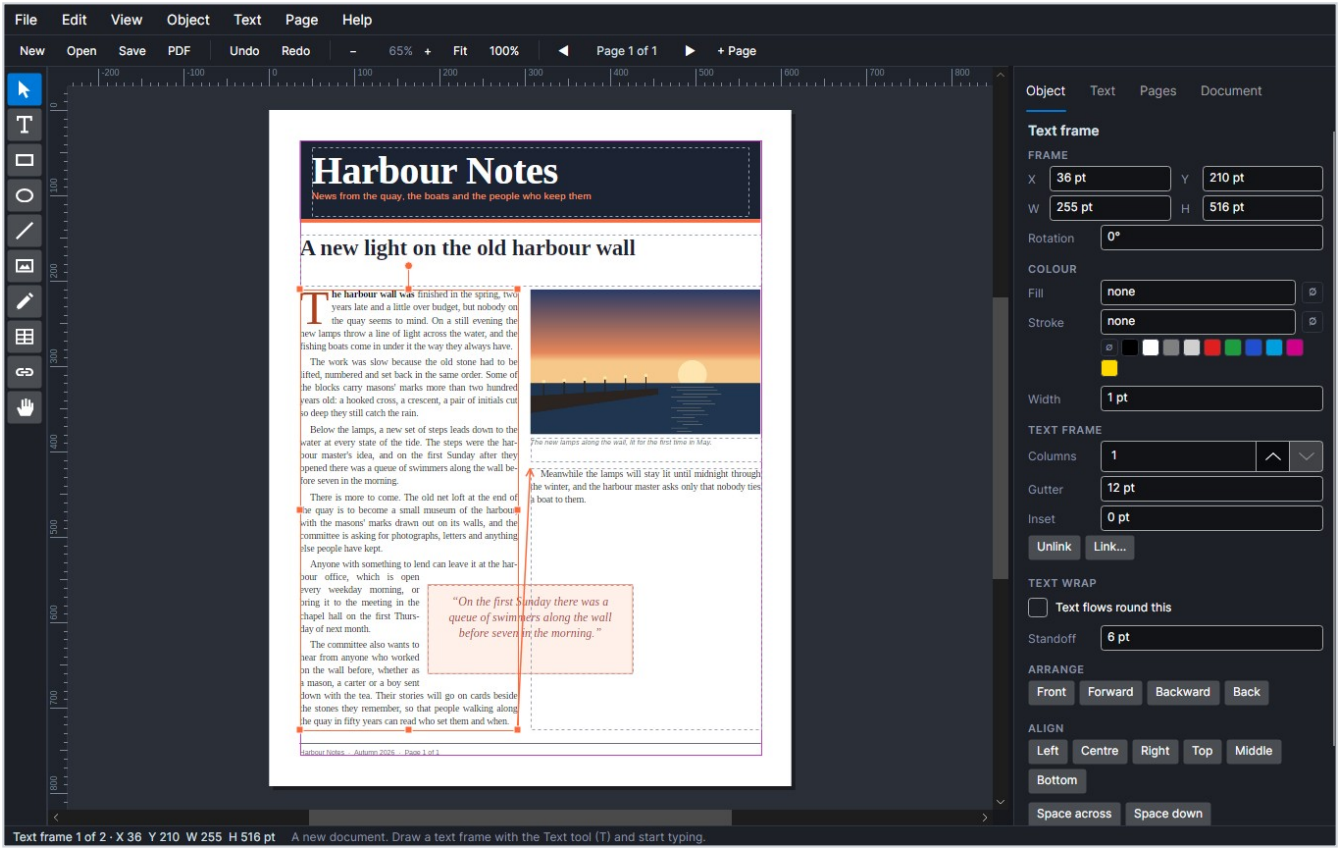
AVAPageStream is a desktop publishing program: newsletters, leaflets, booklets, posters, menus, programmes and anything else that is a page of words and pictures laid out with care. You draw frames on a page, pour text into them and link them so the text flows from frame to frame, then export a PDF or print it.

There is nothing to install. Download the file for your computer from the releases page, unpack it and run it: the .NET runtime and everything else it needs are inside the one file.

- On Linux and macOS the archive keeps the executable bit. If it goes missing on the way, `chmod +x AVAPageStream`.
- macOS will want the file cleared from quarantine before it opens something downloaded from the web.
- To open a document straight away, give it on the command line: `AVAPageStream newsletter.avps`.

Your work is saved in `.avps` files. A document file holds everything, its pictures included, so it opens the same on another computer. Liberation Serif and Liberation Sans are built into the program for the same reason: text set in them looks the same everywhere. You can use the fonts installed on your computer too, but whoever opens the file needs them as well.

The window



Down the left is the **toolbox**: the tools for selecting, typing, drawing and moving about (each has a key, shown when you hover over it). In the middle is the **page**, on a grey pasteboard, with **rulers** along the top and left and scroll bars along the right and bottom. Down the right is the **side panel**, in four tabs:

- **Object**: the selected object's position and size, colours, and everything else about it.
- **Text**: fonts and type, paragraph settings, and paragraph styles.
- **Pages**: the pages and the master pages.
- **Document**: page size, margins, columns, the grid, colour swatches, spelling and units.

The **tool bar** above the page holds the things you reach for most: new, open, save and PDF, undo and redo, zoom, and moving between pages. The **status bar** along the bottom says what is selected (with its position and size), what just happened, and where the mouse is on the page.

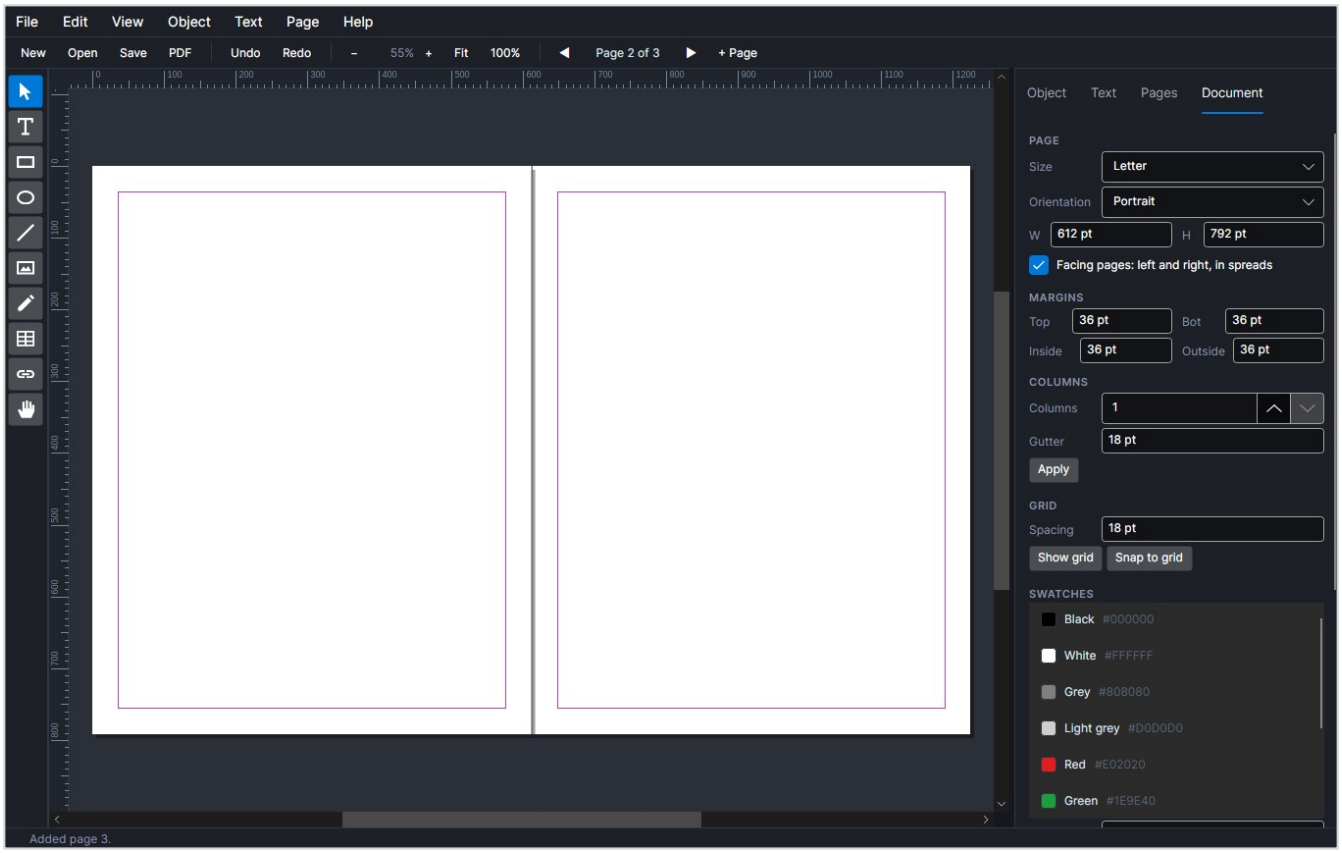
The pasteboard is yours to work on. Leave things half on and half off the page while you decide; only what is on the page is printed or exported.

Moving about. The mouse wheel scrolls (Shift+wheel sideways), Ctrl+wheel zooms in and out about the mouse, Ctrl+0 fits the page in the window and Ctrl+1 shows it at actual size. Hold Space, or use the Hand tool (H), and drag to slide the page about. Page Up and Page Down go to the previous and next page.

Measures can be typed in any unit, whatever the document's unit is: `54`, `0.75in`, `20mm`, `2.5cm`, `4p6` (4

picas and 6 points) or `18pt`, and simple sums such as `1in + 6pt`. A box shows the measure back in the document's unit, which you choose at the foot of the Document tab.

Setting up a document

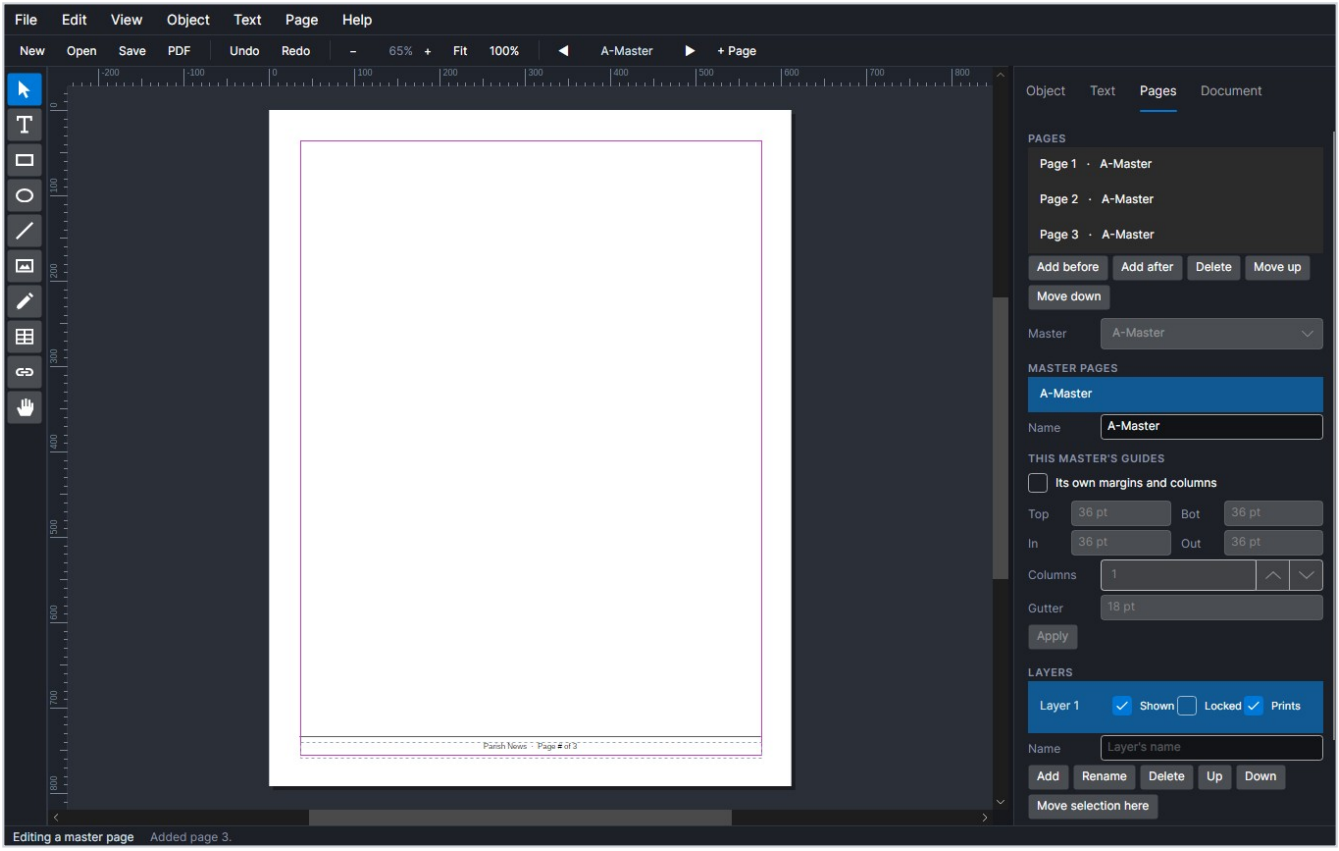


A new document is one Letter page with half-inch margins and one column. Change it in the **Document** tab:

- **Size**: Letter, Legal, Tabloid, A3, A4, A5, B5, or type your own width and height. **Orientation** turns it.
- **Margins** and **Columns** (with the **Gutter** between them) are the guides drawn on every page: margins in magenta, columns in violet. They are only guides, for snapping to; text frames are what text flows in.
- **Apply** puts the size, margins and columns into effect, as one step you can undo.

Tick **Facing pages** for anything that will be printed both sides and bound: a booklet, a magazine, a book. Odd pages are then right-hand pages and even pages left-hand ones, page 1 on its own on the right, the way a printed book opens. The view shows a **spread**, the left and right pages side by side with the spine between, and you work on either by clicking it. The left and right margins become the **inside** (by the spine) and the **outside**, so a left page has them the other way round.

Pages and master pages



The **Pages** tab lists the pages. **Add before** and **Add after** put a page in beside the one shown, **Delete** takes the one shown out (never the last), and **Move up** and **Move down**, or dragging a page in the list, change the order. Click a page to show it.

Every page uses a **master page**, "A-Master" to start with. Whatever is on a master is drawn under every page that uses it: a running head, a footer, a rule, a logo, a page number. Pick a master in the list of master pages to edit it (or Page › Edit this page's master); what you draw there appears on its pages, which can't select it. Choose a page's master under the page list, or none.

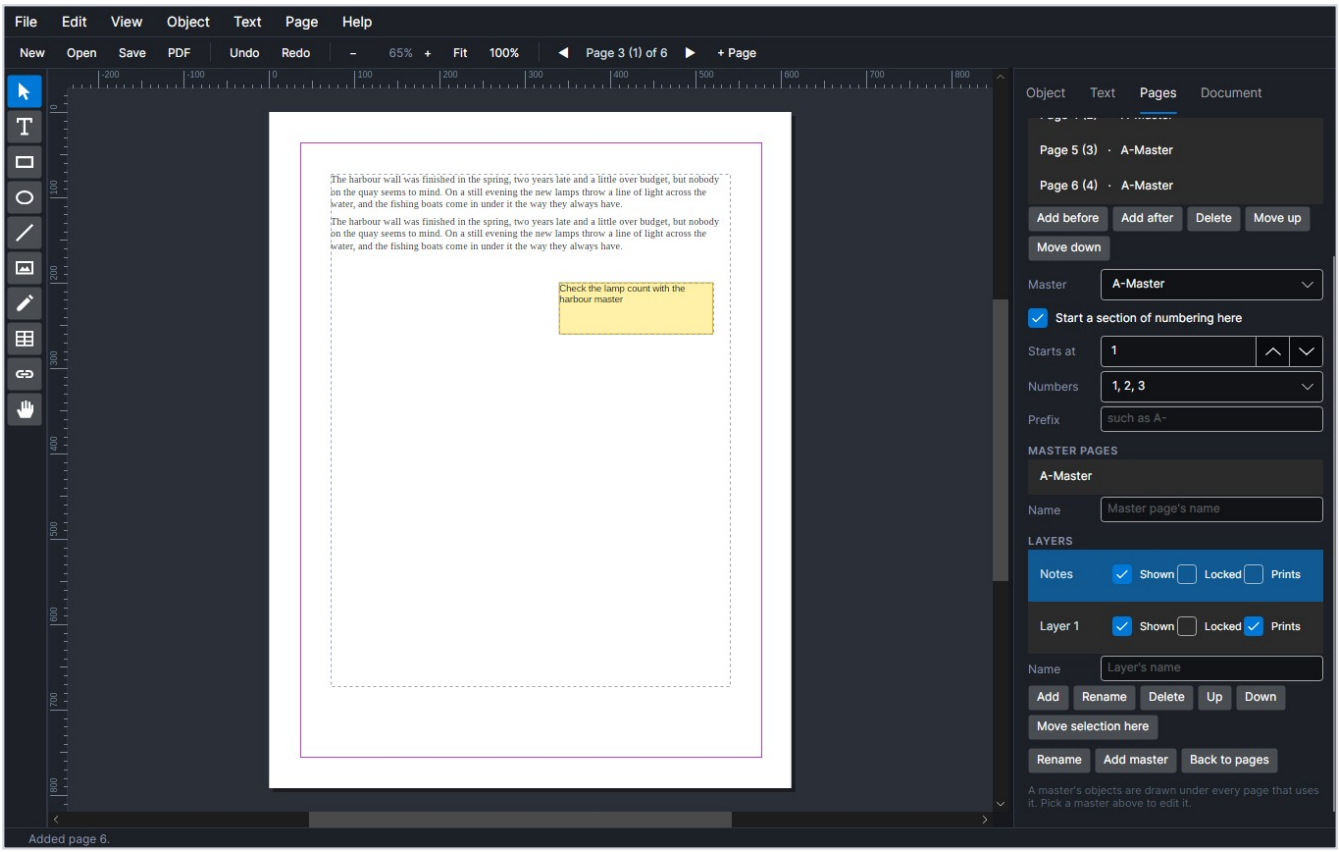
Page numbers. Put a text frame on the master, type the words round it, and choose Text › Insert page number (Ctrl+Shift+3). On the master it shows as "#", and on each page that page's own number. Text › Insert page count adds the number of pages, for "Page 3 of 12". A page number in a frame on a page itself counts from the page the story starts on.

Sections of numbering. Front matter numbered i, ii, iii and the chapters from 1, or an appendix numbered A-1, A-2: show the page a section starts on, and in the Pages tab tick **Start a section of numbering here**. Then set what its first page is numbered (**Starts at**), how (**Numbers**: 1, 2, 3; i, ii, iii; I, II, III; a, b, c or A, B, C) and a **Prefix**. Every page until the next section counts on from there. Page numbers, the page list, the status bar ("Page 5 (iii) of 40") and a table of contents all use the section's numbers.

A master can have **its own margins and columns** (This master's guides, in the Pages tab), for the pages

that use it: an index set in three columns, say, in a document of two. In a facing document each master has a left side and a right side, drawn under left and right pages, so page numbers can sit on the outside corners of both.

Layers

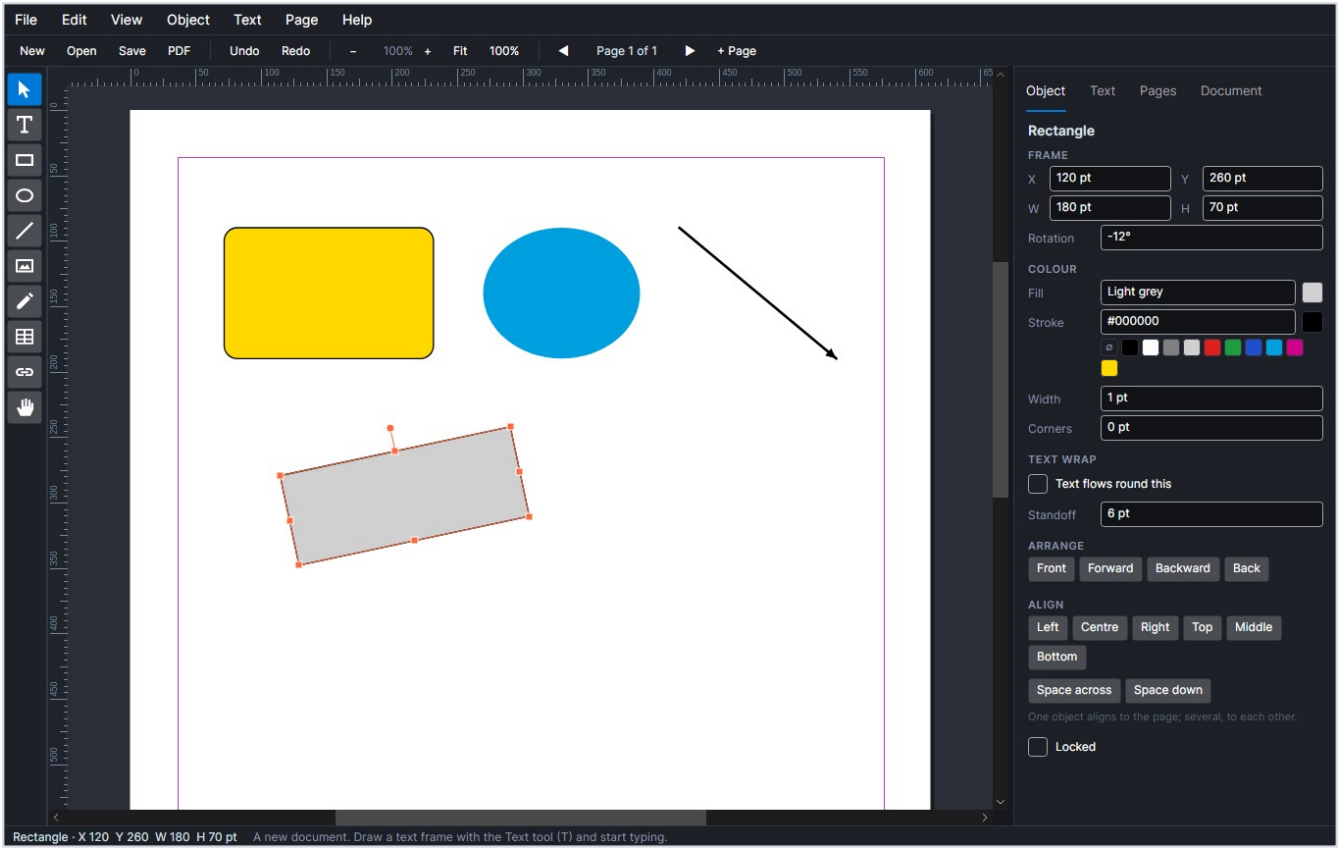


The Pages tab's **Layers** list holds the document's layers, top first. Every object is on one, and the layers are drawn in order, so a layer of notes or a watermark can sit above or below everything else whatever order things were drawn in. Each layer has three ticks:

- **Shown**: untick it to hide the layer's objects, on screen and in print. A hidden layer's objects don't wrap text.
- **Locked**: its objects are drawn but can't be clicked, dragged or selected, so artwork can't be nudged by mistake.
- **Prints**: untick it for a layer of notes or guides you want to see but not print or export.

Click a layer to make it the one new objects go on. Type a name and click **Add** for a new layer, or **Rename**; **Delete** puts the layer's objects onto the layer below it (the last layer can't be deleted). **Up** and **Down** change the order, and **Move selection here** moves the selected objects onto the picked layer.

Drawing



Tool	Key	
Select	V	Click to select, Shift+click to add or take away, drag across the page to select everything inside.
Text	T	Drag a text frame and type in it; click in a frame to put the caret there.
Rectangle	R	Drag to draw. Shift makes a square.
Ellipse	E	Drag to draw. Shift makes a circle.
Line	L	Drag to draw. Shift keeps it to 45° steps.
Picture	P	Drag a frame, then pick a picture for it.

Pen	B	Polygons and curves: see below.
Table	G	Drag a table: see Tables.
Link	K	Link text frames so text flows from one to the next.
Hand	H	Drag to slide the page about (or hold Space with any tool).

Moving and resizing. Drag an object to move it, or one of the eight handles round it to resize it (Shift keeps its shape). The arrow keys nudge the selection by a point, ten with Shift. Moving and resizing snap to the margins, columns, guides and the page's edges and middle (see Rulers, guides and the grid). Or type exact figures into **X**, **Y**, **W** and **H** in the Object tab, in any unit.

Colour. Fill and stroke are set in the Object tab: type a colour's name, a value like `#2050D0`, or `none`, or click a swatch below them (Shift+click for the stroke). **Width** is the stroke's thickness. A new object is drawn with the fill and stroke last set on one of its kind. Rectangles have rounded **Corners**; lines have **Arrowheads** at either end.

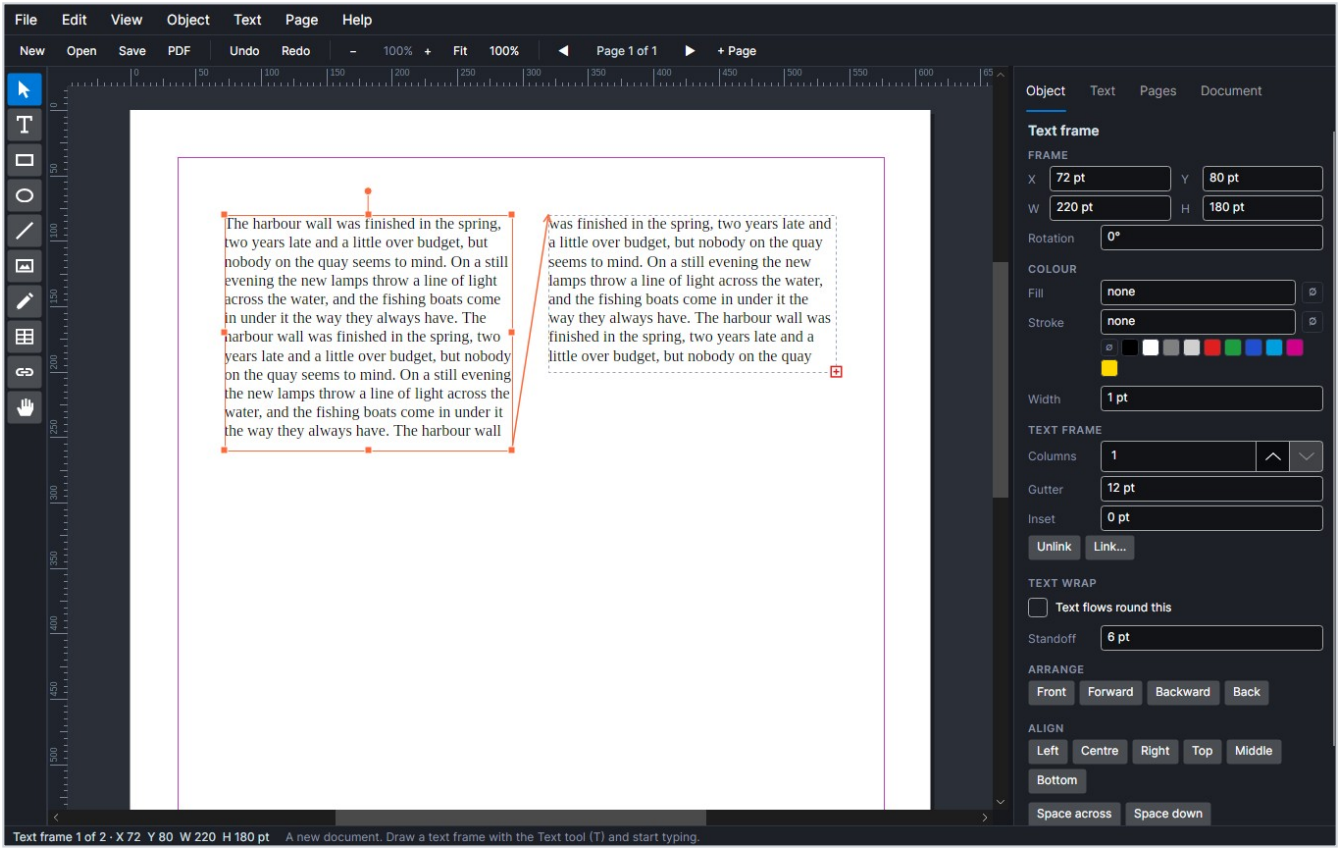
Turning. Drag the round handle above a selected object to turn it about its middle (Shift for 15° steps), or type the angle into **Rotation**. A turned text frame still takes typing and clicks the right way up for it.

Arranging. Objects are drawn in order, the last on top. Object › Bring to front and Send to back, or Bring forward and Send backward (Ctrl+] and Ctrl+[, with Shift for all the way), change the order. The Object tab's **Align** buttons line selected objects up by their edges or middles (one object, with the page); **Space across** and **Space down** spread three or more evenly.

Grouping. Ctrl+G makes the selected objects one group, which moves, resizes and arranges as one; Ctrl+Shift+G takes it apart. You can still type in a text frame inside a group: double-click it.

Copying and deleting. Ctrl+D duplicates the selection a little down and to the right; Ctrl+C and Ctrl+V copy and paste it, on this page or another; Delete takes it away. **Lock** (in the Object tab or menu) keeps an object from being moved, resized or deleted by accident.

Text frames and how text flows



Text lives in **stories**. A story flows through one text frame, or through a chain of them linked together, column by column and frame by frame, and reflows whenever you type, resize a frame or change the type.

Draw a text frame with the Text tool and start typing. When there is more text than the frame holds, a small red square with a plus in it appears at its bottom-right corner and the status bar says how many characters are **overset**: there, but not showing. To give the text somewhere to go:

1. Draw another text frame, empty, on this page or any other.
2. Click the red mark with the Select tool (or pick the Link tool, K, and click the first frame).
3. Click the new frame. The text flows on into it.

When a linked frame is selected, arrows show the way the text goes from frame to frame. Link as many frames as you like; a story can run through a whole book. **Unlink** in the Object tab (or Alt+click with the Link tool) takes a frame out of its chain, and the text flows on through the rest.

A frame can have several **Columns** of its own, with a **Gutter** between them, and an **Inset** that keeps the text in from its edges (Object tab › Text frame).

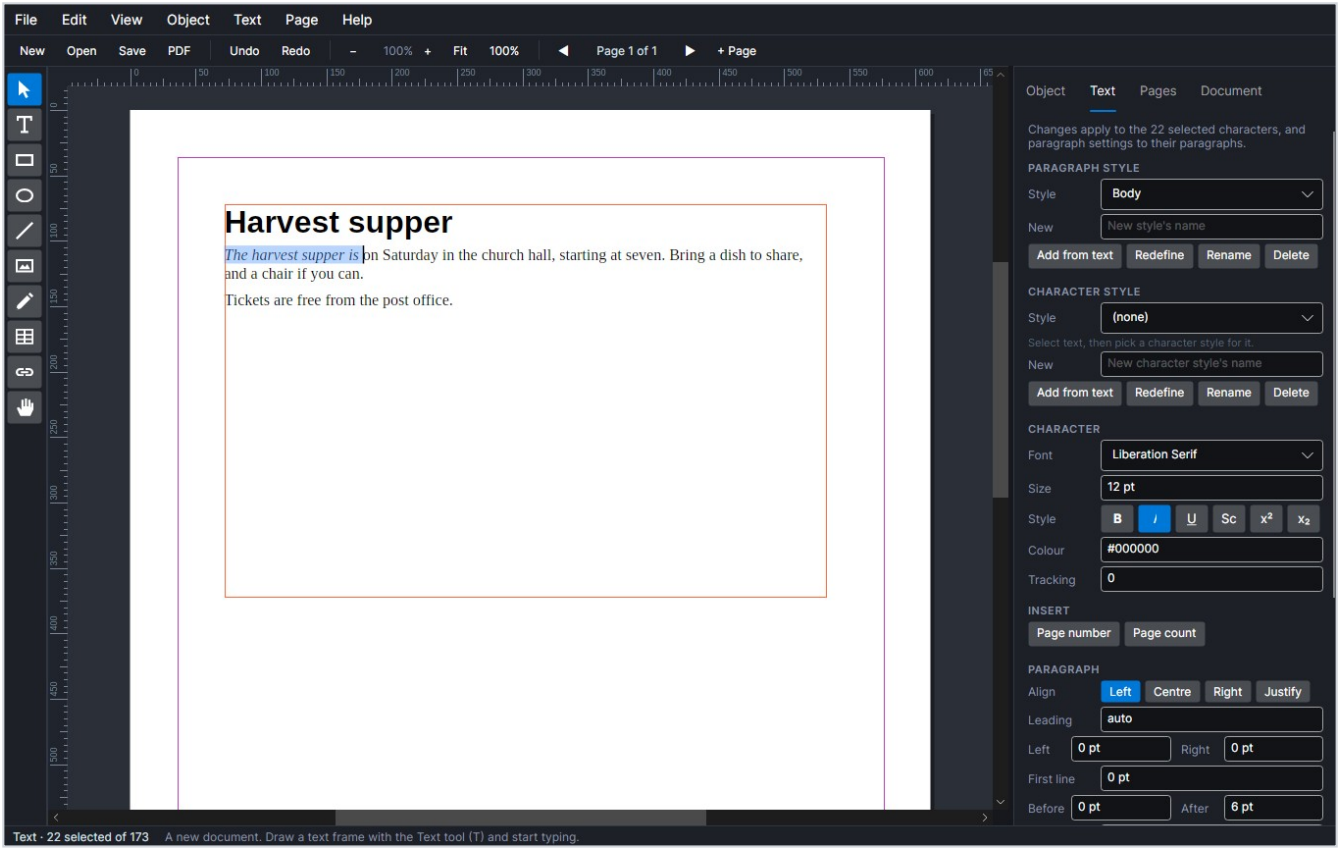
Typing and editing

Double-click a text frame with Select, or click in it with the Text tool, to put the caret in it. The caret moves through the whole story, from frame to frame and page to page; the view follows it.

Keys	
Arrows, Home, End	Move by a character or line, to the start or end of the line.
Ctrl+Left, Ctrl+Right	Move by a word.
Ctrl+Home, Ctrl+End	The start or end of the story.
Page Up, Page Down	The start of the previous or next frame in the chain.
Shift with any of those	Select as you go.
Double-click, triple-click	Select a word, a paragraph.
Ctrl+A	Select the whole story.
Backspace, Delete	Delete a character (with Ctrl, a word).
Enter	A new paragraph.
Escape	Leave the text; the frame stays selected.

Cut, copy and paste work on the text as you would expect. Text copied and pasted within AVAPageStream keeps its fonts, sizes and styles; text from other programs comes in plain, taking the look of where it lands.

Formatting text



The **Text** tab sets the type of what is selected. With the caret in some text, its character settings change the selected characters (or, with nothing selected, the next you type) and its paragraph settings change the paragraphs the selection touches. With text frames selected and no caret, they change all the frames' text.

- **Font**, **Size**, and **B**, **I** and **U** (Ctrl+B, Ctrl+I, Ctrl+U) for bold, italic and underline.
- **Sc** sets small capitals; **x²** and **x₂** superscript and subscript.
- **Colour** of the text, by a swatch's name or a value.
- **Tracking** loosens or tightens the spacing, in thousandths of an em. To kern a pair of letters by hand, put the caret between them and press Alt+Left or Alt+Right (Shift for bigger steps).
- **Align**: left, centre, right or justified.
- **Leading**: the distance from one line to the next, or `auto` (120% of the largest type on the line).
- **Left** and **Right** indents, the **First line** indent (a negative one makes a hanging indent), and space **Before** and **After** the paragraph.

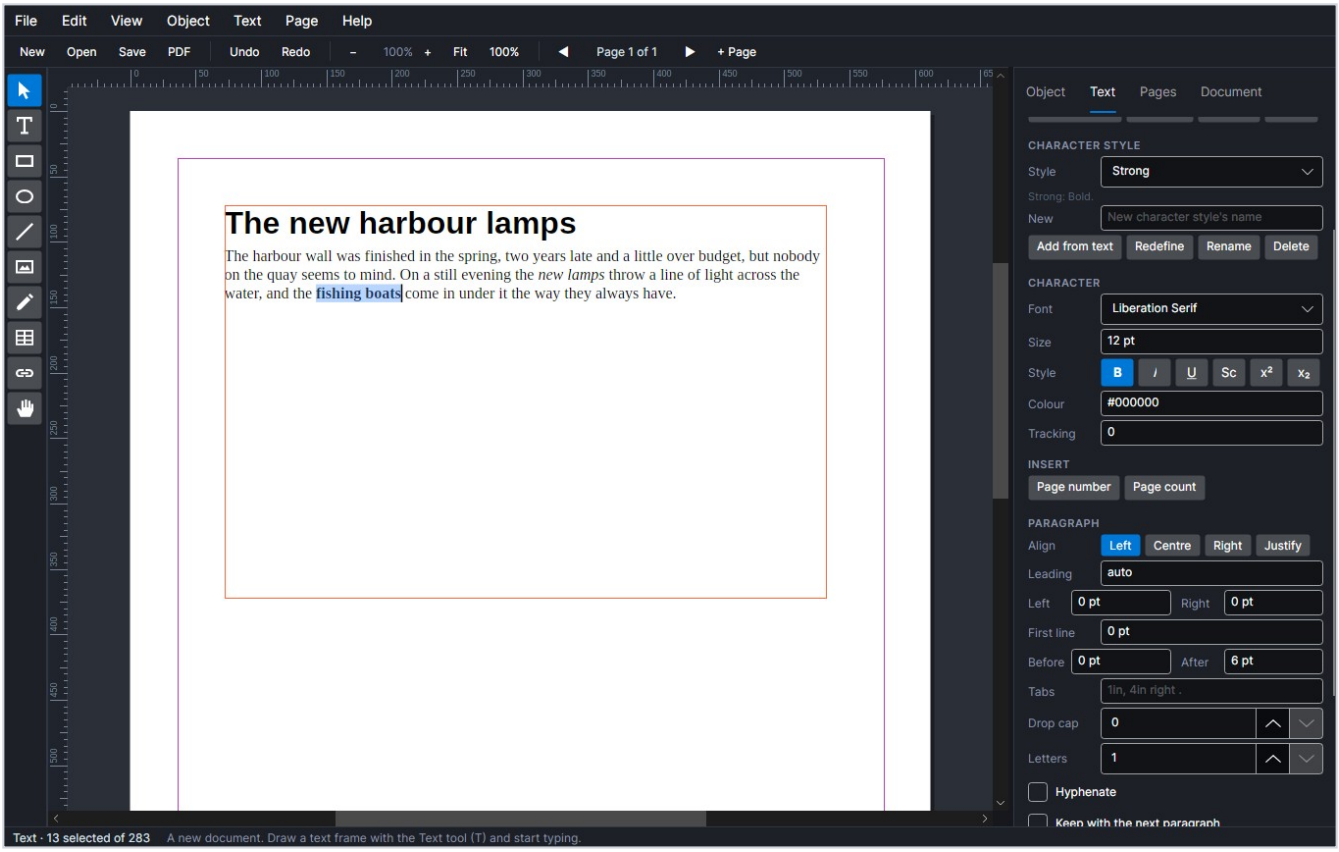
Paragraph styles

A paragraph style is a named set of all of that: Body, Heading, Subheading and Caption to start with. Pick one from **Style** to set the paragraph the caret is in. Styles are what keep a long document consistent, and

what make it quick to change: to make every heading a size bigger, change one heading, then **Redefine**, and they all follow.

- **Add from text**: type a name and click it, and the paragraph the caret is in becomes a new style.
- **Rename** and **Delete** do as they say; deleting a style leaves its paragraphs looking as they did.

Character styles



A character style is a named set of a few character settings, laid over whatever paragraph style the text is in: **Emphasis** (italic), **Strong** (bold) and **Small caps** to start with. Select some words and pick one from the character **Style** list; pick **(none)** to take it off again. Because it holds only the settings it changes, Emphasis in a heading is italic heading type, and in body text italic body type, and changing the paragraph's style keeps it.

- **Add from text**: format some words the way you want, type a name and click it. The new style is what those words have that their paragraph's style doesn't, shown beneath the list.
- **Redefine** makes the style what the selected words are now, and every word using it follows.
- **Rename** and **Delete**; deleting a character style puts its words back to their paragraph's type.

Placed Markdown uses Strong for `**bold**` and Emphasis for `*italic*`.

Keeping text together

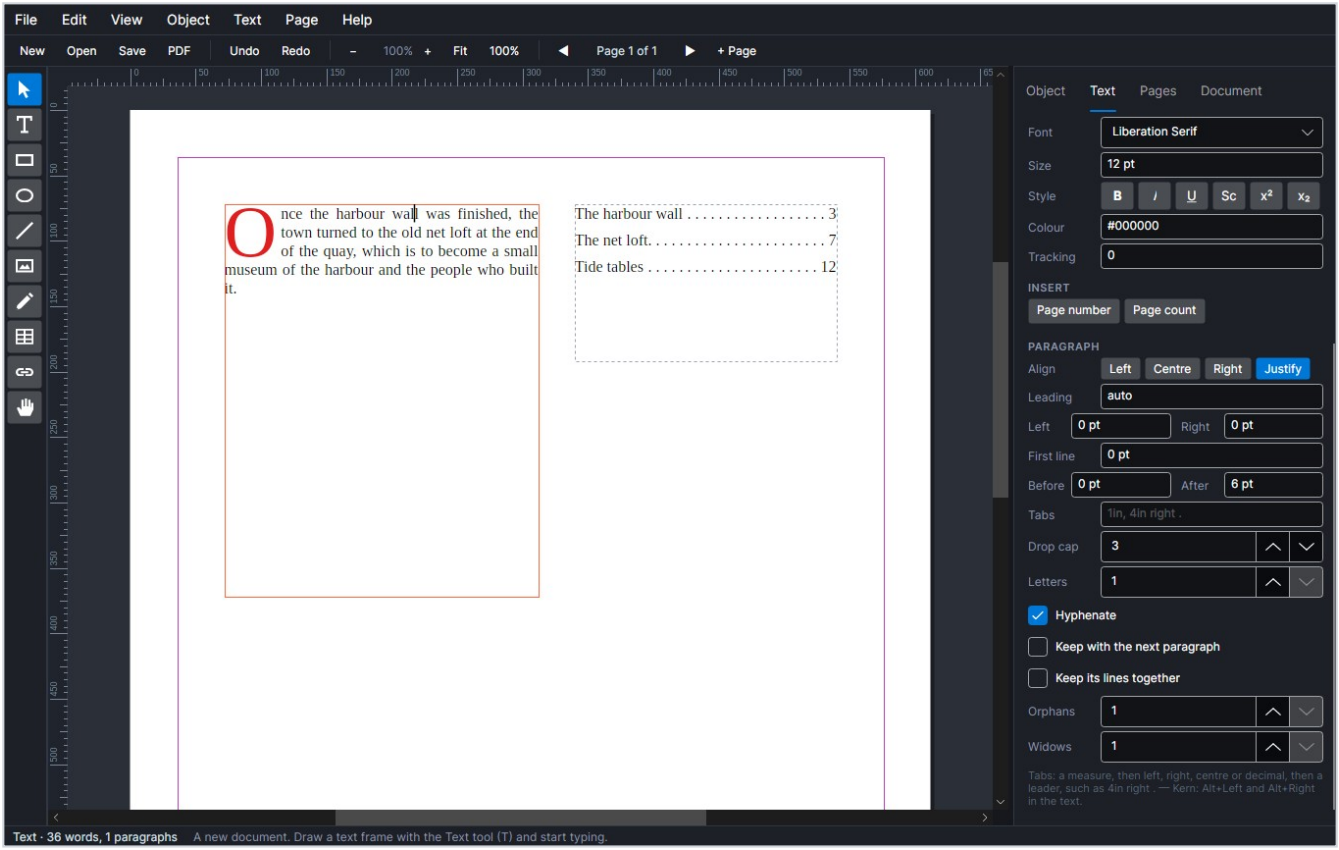
Four settings at the foot of the Text tab stop awkward breaks between columns and pages, for the

paragraph the caret is in (or for a paragraph style, by redefining it):

- **Keep with the next paragraph**: a heading never sits alone at the foot of a column; it goes on with the paragraph after it.
- **Keep its lines together**: the paragraph never breaks; if it doesn't fit, it all starts the next column.
- **Orphans**: at least so many of its lines at the foot of a column, or none there.
- **Widows**: at least so many of its lines at the top of the next, brought over from the column before if need be.

A paragraph already at the top of a column, or too long for any column, is never moved, so text can't vanish.

Tabs, drop caps and hyphenation



Tabs. Press Tab in the text and it jumps to the next tab stop: every half inch, unless the paragraph has stops of its own. Type them in the Text tab's **Tabs** box, separated by commas: a measure, then left, right, centre or decimal, then a leader character to fill the gap if you want one. `1in, 4in right .` puts a left tab at an inch and a right tab at four inches with dots leading to it, as in a list of contents. Decimal tabs line figures up on their decimal points. A hanging indent's left indent is a stop too, so a bullet, a tab and the text line up.

Drop caps. Drop cap sets how many lines deep it goes (0 or 1 for none), and **Letters** how many letters it takes. The capital stands from the last of those lines' baseline up to the top of the first line's capitals, and the lines beside it make room.

Hyphenation. Tick **Hyphenate** and words are broken at the ends of lines where the hyphenation patterns allow, with at least two letters before the break and three after it. Justified text looks far better with it.

Placing text

Text written elsewhere doesn't need typing again. Select a text frame (or put the caret where the text should go) and choose Object › Place text…, then pick a `.txt` or `.md` file.

- Plain text comes in a paragraph a line, in the Body style.
- **Markdown** comes in styled: `#` headings take the Heading style and `##` and smaller Subheading, paragraphs take Body, list items get a bullet or number on a hanging indent, quotes are indented and in italic, and `**bold**` and `*italic*` carry over.

What doesn't fit is overset, with the red mark showing: link more frames to flow it on.

Pictures

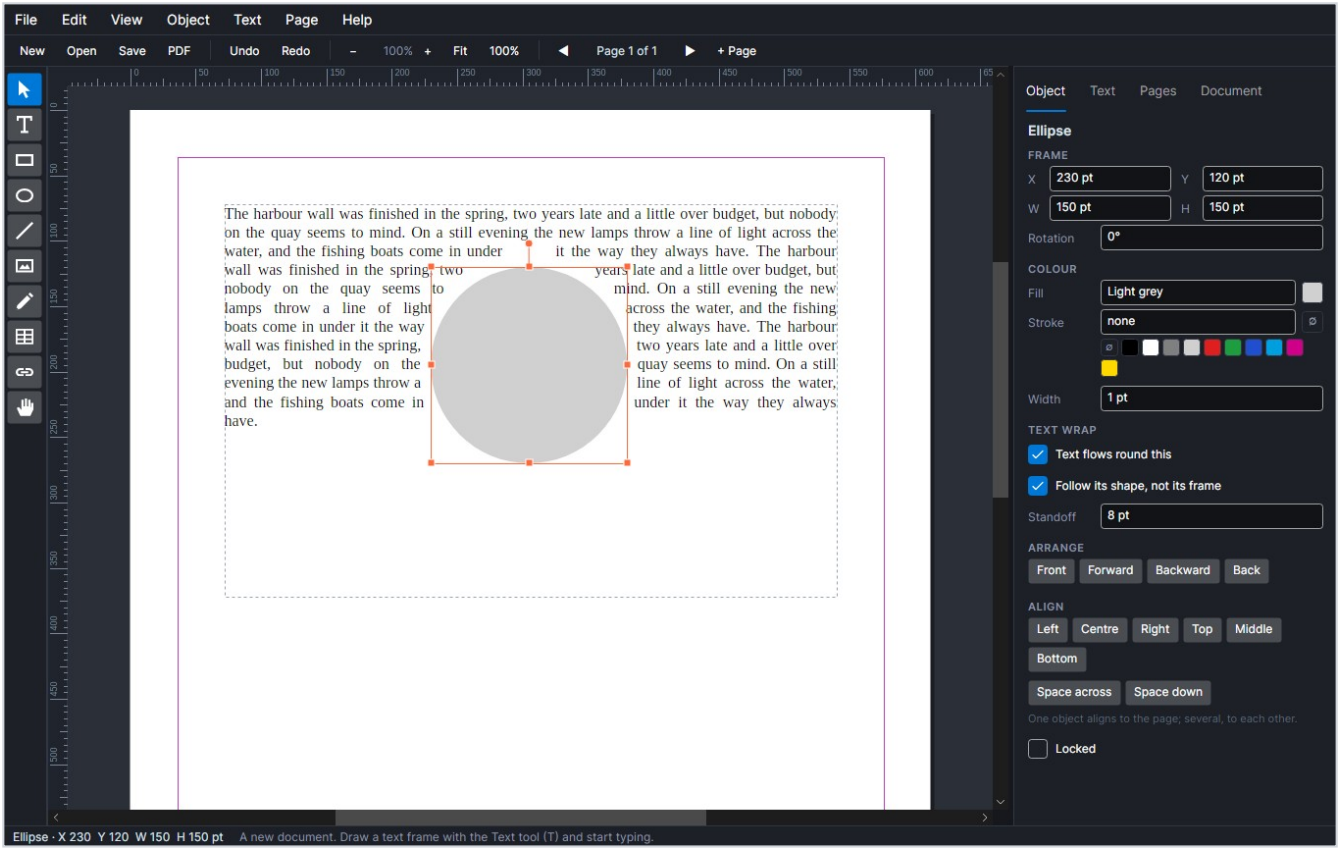
Draw a frame with the Picture tool and pick a picture, or select a picture frame and choose Object › Place picture…. Click a picture with the Picture tool to replace it. Pictures can be:

- **PNG, JPEG, GIF, BMP and WebP**: the usual kinds.
- **SVG** drawings: kept as drawings, so they stay sharp at any size, on screen and in PDFs.
- **EPS** (Encapsulated PostScript): sized from the file. With Ghostscript installed it is shown and printed in full; without it, by the preview many EPS files carry, or as a labelled box.

The picture is stored inside the document, so moving or deleting the original file does no harm.

Fit (Object tab) decides how a picture fills its frame: **Fit** shows all of it, **Fill** fills the frame and crops what spills over, **Stretch** fills it out of shape, and **Actual** is the picture's own size. To crop a different part into view, Alt+drag the picture with Select, or type an **Offset**.

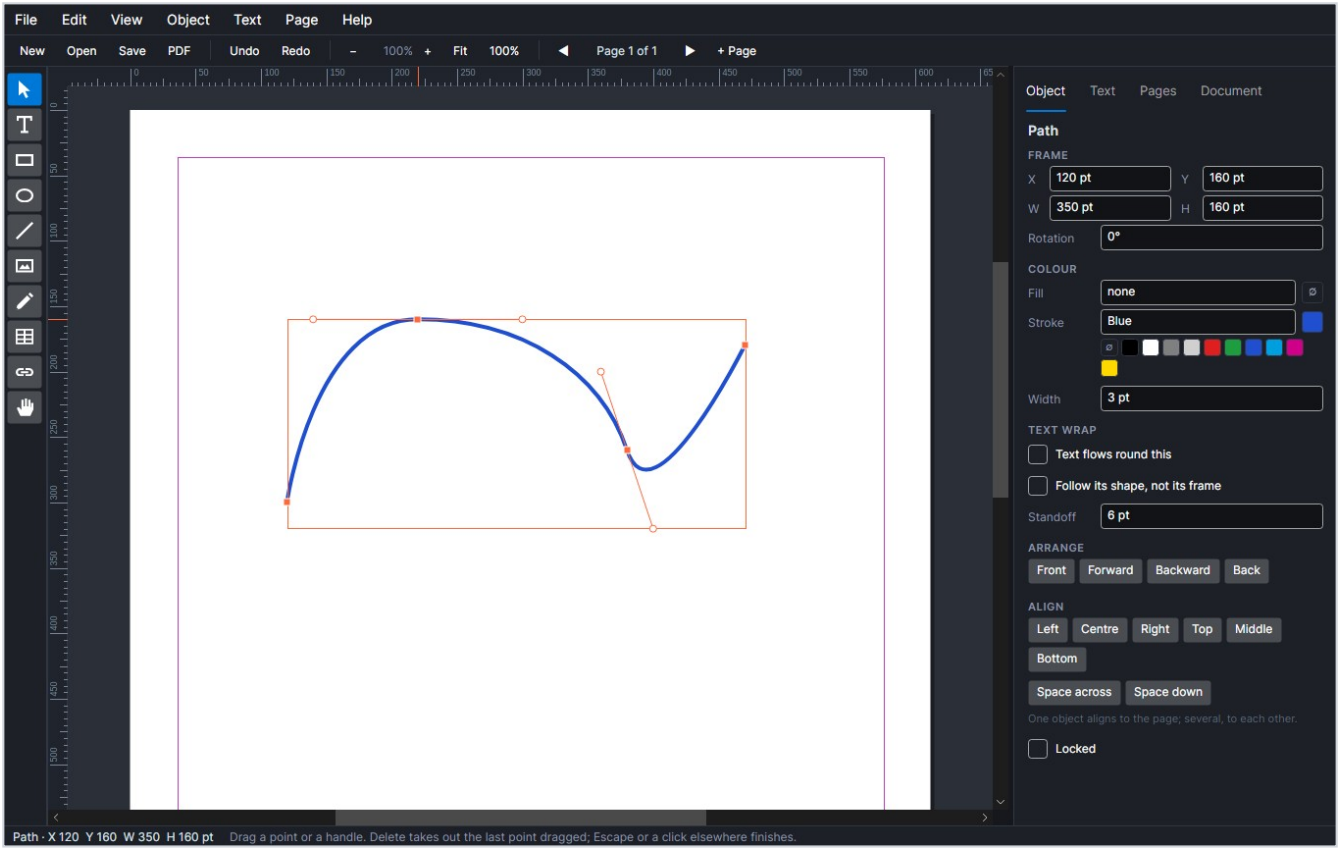
Text wrap



Tick **Text flows round this** for any object (a picture, a box, a frame, a shape) and the text in the frames under it keeps out of its way, the **Standoff** from its edge. Where there is room on both sides, the text runs on either side; where there isn't, it moves on below. For an ellipse or a closed path, **Follow its shape** makes the text hug its outline rather than its frame.

Wrap affects only the text **under** the object, in the drawing order. So a pull quote's own text frame, sitting on top of its tinted box, keeps its text, while the columns underneath flow round both. Use Bring forward and Send backward to decide which is which. Objects on a master page wrap the text on every page that uses it.

Polygons and curves

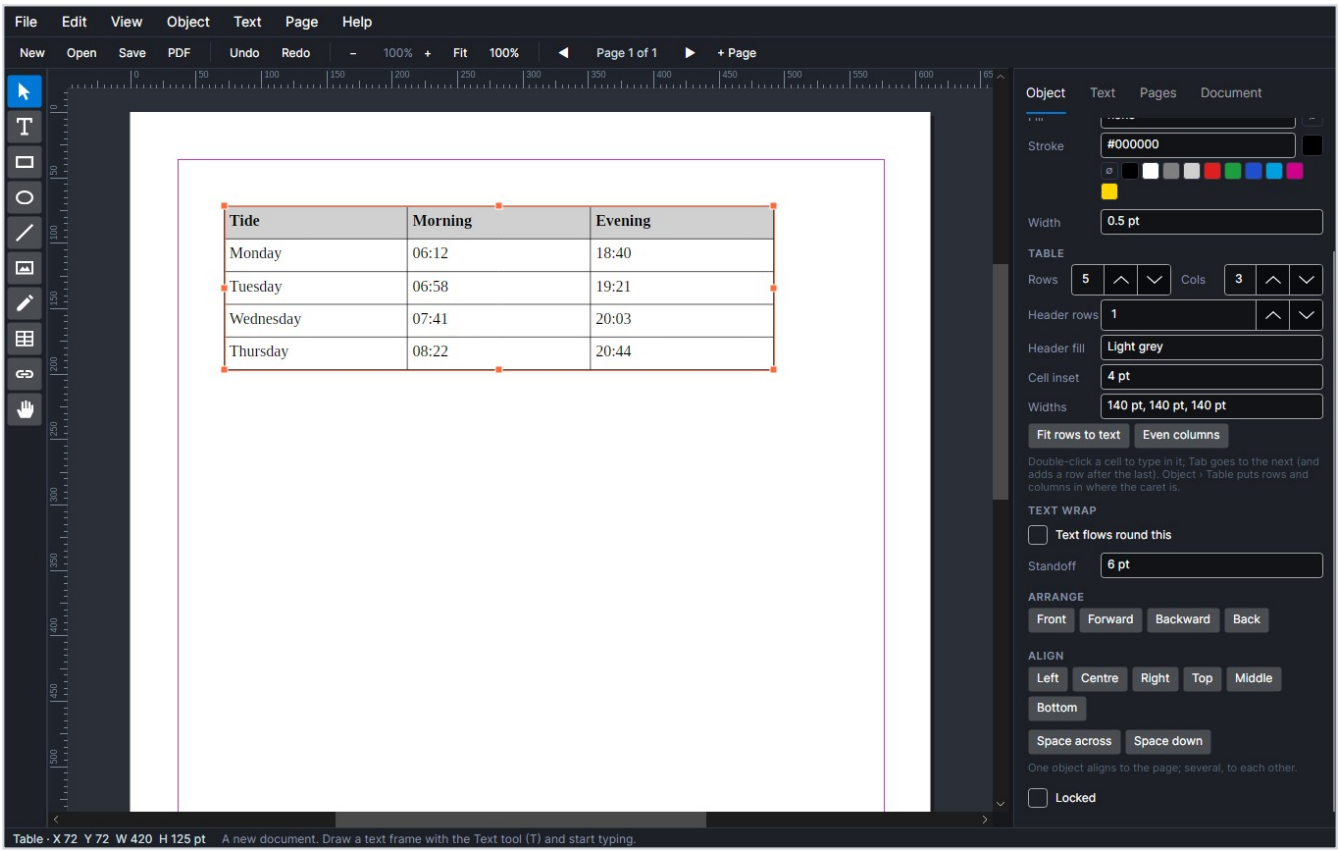


The **Pen** (B) draws shapes of your own, point by point:

- **Click** for a corner point, and **drag** for a smooth one: the handle comes out to where you let go and sets the curve through the point. Shift keeps a point to 45° steps from the last.
- **Click the first point** again to close the shape (only a closed shape is filled).
- **Double-click**, or press Enter, to finish an open path; Escape gives up the one you're drawing.

Double-click a path with Select to change it: drag its points (squares) and their handles (dots). Delete takes out the last point you dragged. Escape, or a click elsewhere, finishes.

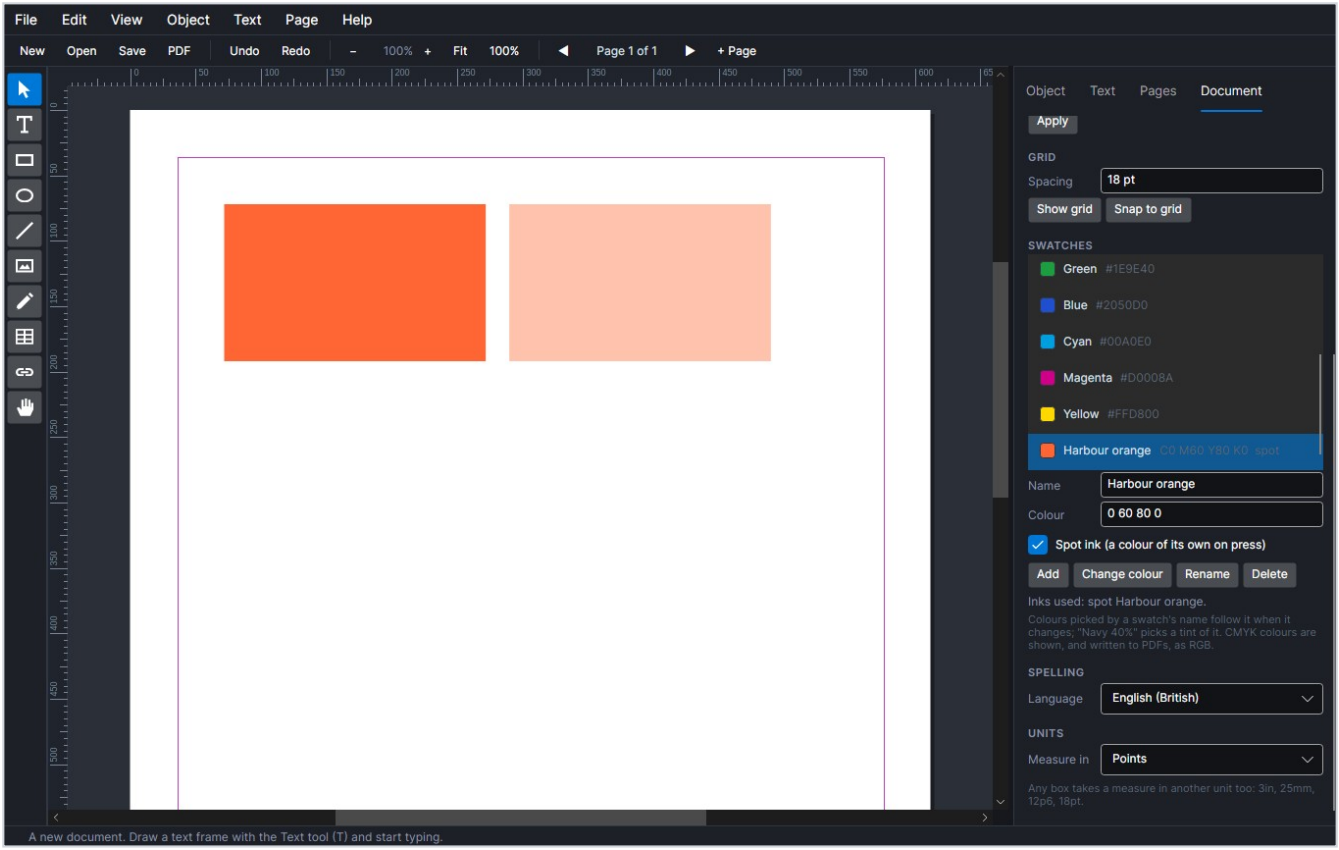
Tables



Draw a table with the **Table** tool (G): it starts with four rows and three columns, the first a shaded header row, and the caret in the first cell. Each cell holds its own text, formatted like any other.

- **Tab** moves to the next cell (Shift+Tab back). Tab in the last cell adds a row.
- Double-click a cell (or click it with the Text tool) to type in it, and click another cell to move there.
- The Object tab sets **Rows** and **Cols** (added or taken from the end), **Header rows** and their **Header fill**, the **Cell inset**, and the **Widths** of the columns (`1in, 2in, 1in`). **Even columns** shares the width out.
- Object › Table puts a row in above or below the caret's cell, or a column to its left or right, and deletes the caret's row or column.
- Rows keep the height they have. A cell whose text doesn't fit gets a red mark in its corner: **Fit rows to text** makes each row as tall as its fullest cell needs.
- The table's stroke draws the rules between and round the cells. Resizing the table scales its rows and columns.

Colour and swatches

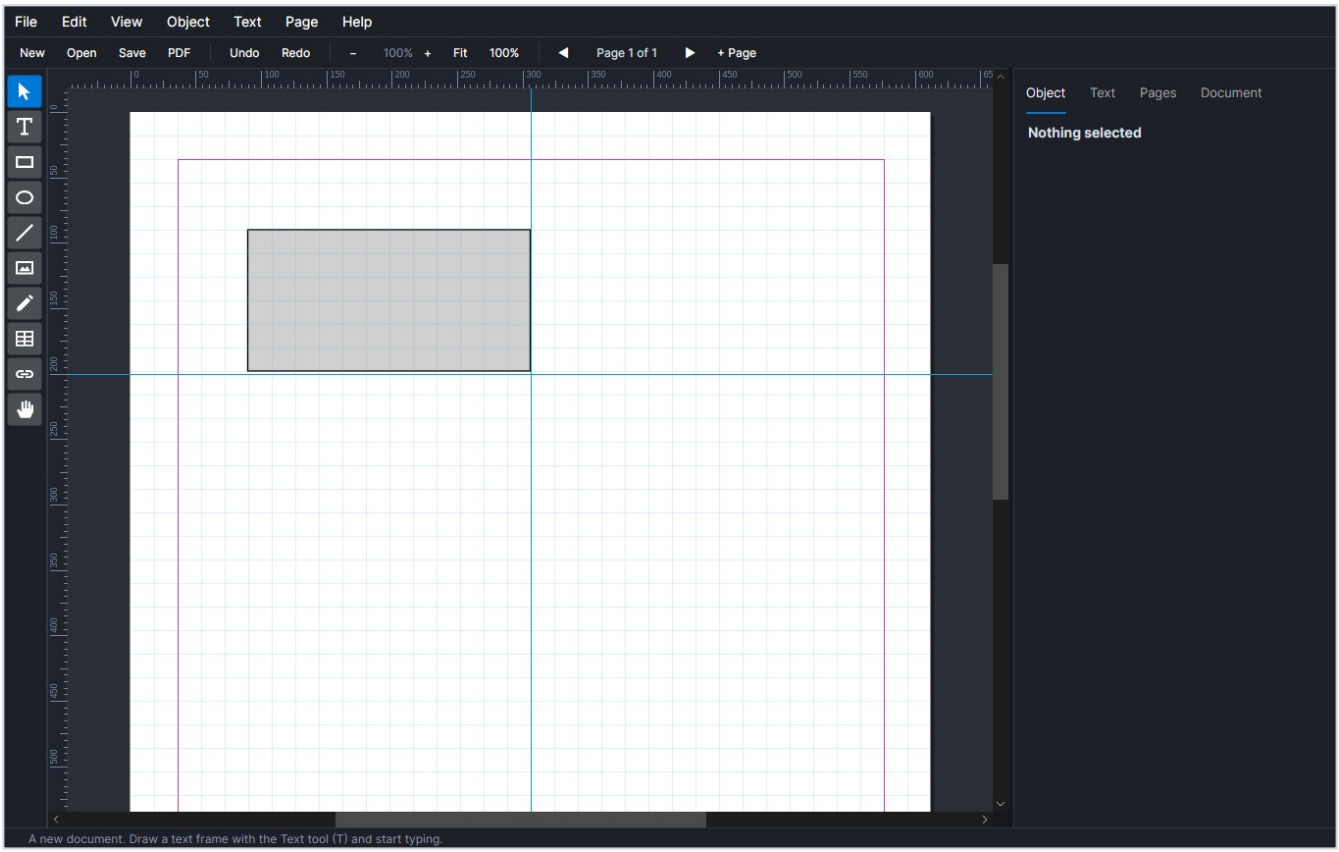


The document keeps a list of named colours, its **swatches** (Document tab › Swatches). Pick a colour by a swatch's name, by typing it or clicking a swatch button, and it stays tied to the swatch: change the swatch's colour and everything using it changes. Add **40%** (or any per cent) after the name for a lighter **tint** of it: "Navy 40%".

- **Add** makes a swatch from a name and a colour. The colour can be `#RRGGBB`, or **CMYK** inks in per cents, such as `0 60 80 0` (cyan, magenta, yellow, black).
- Tick **Spot ink** for a colour printed as an ink of its own, such as a Pantone colour.
- **Change colour**, **Rename** and **Delete** do as they say. Renaming takes everything using the swatch with it; deleting one leaves its colour where it was used.
- **Inks used** tells you what the document needs on press: process colour (CMYK) and the name of each spot ink.

CMYK colours are shown on screen by a plain conversion, and PDFs are written in RGB. For exact print colour, tell your printer which spot inks and CMYK values you mean.

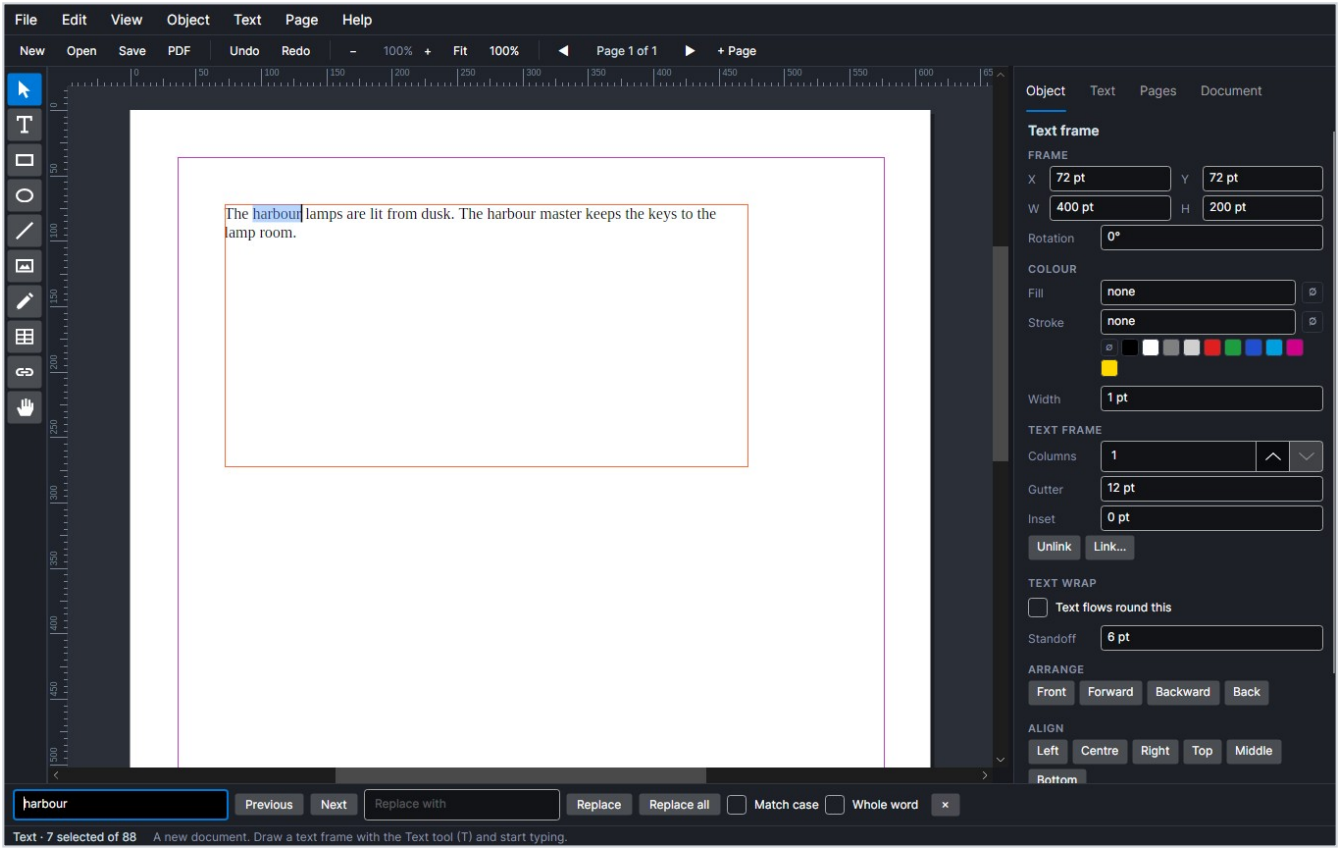
Rulers, guides and the grid



The **rulers** measure from the top-left corner of the page you are working on, in the document's unit, with a mark following the mouse.

- Drag from a ruler onto the page to make a **guide**: from the top ruler a level one, from the left an upright one. Drag a guide to move it, and back onto its ruler to remove it. Guides belong to their page; a master's guides show on its pages.
- **Snap** (View › Snap to guides) makes moving, resizing and drawing land on the margins, column edges, guides and the page's edges and middle, when within a few pixels of them.
- View › **Show grid** draws the document grid, and **Snap to grid** makes things land on it too. Set its spacing in the Document tab (18 pt to start with).
- View › Show guides hides and shows them all.

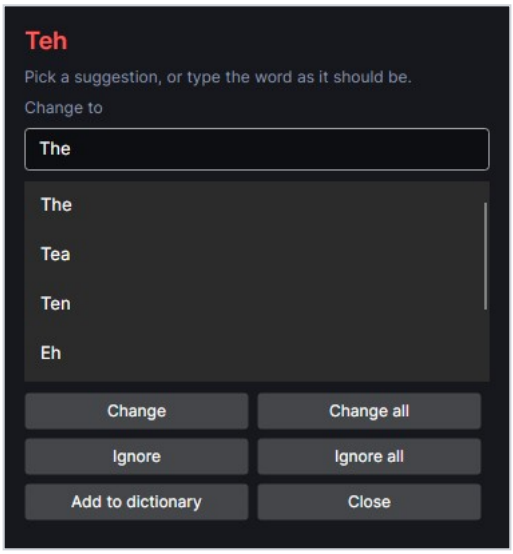
Find and replace



Ctrl+F opens the find bar under the page. Type what to find and press Enter (or F3) for the next, Shift+Enter for the one before. It looks through every story in reading order, page by page and then the master pages, and shows each one selected on its page, with how many there are. **Match case** and **Whole word** narrow it down.

Replace changes the one found and finds the next; **Replace all** changes every one at once, as a single step to undo. Each replacement takes the look of the text it replaces. Escape closes the bar.

Spelling



Misspelt words are underlined in red as you type (on screen only, never in print). View › Check spelling as you type turns that off. The Document tab's **Spelling** sets the language: American or British English.

Text › Check spelling (F7) goes through the document from the caret, word by misspelt word, showing each on its page:

- **Change** puts in the suggestion picked (or what you type in Change to); **Change all** changes it everywhere.
- **Ignore** passes this one; **Ignore all** passes the word everywhere in this document (and remembers it with the document).
- **Add to dictionary** puts the word in your own dictionary, for every document: names, places, words of your trade.

Capitalised words and words in capitals are checked as the dictionary has them in lower case, and a name keeps its capital: "London" is right, "london" isn't.

The story editor



For writing and correcting without the layout in the way, Text › Story editor (Ctrl+Shift+E) opens the story of the selected frame (or the one you're typing in) as plain text in a window of its own. Type there and the page changes as you go, the formatting round your changes kept; change the page and it shows there. The paragraph style of the paragraph you're in is shown at the top, to see and to set, with the story's word count and how much is overset.

A table of contents

Put the caret where the contents should go (or select an empty text frame) and choose Text › Table of contents…. Tick the paragraph styles to list, the first ticked as the top level: Heading and Subheading, usually. Each paragraph in those styles becomes a line of the contents, with its page number at the right after a row of dots. The contents are set in styles of their own, Contents 1, Contents 2 and so on, which you can change like any other.

The page numbers are the pages as they are now. After the pages move about, make the contents again.

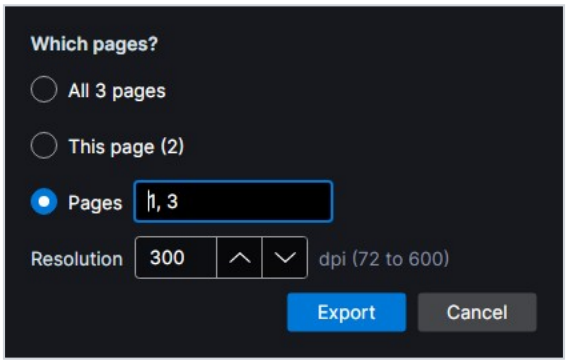
Templates

A template is a document to start others from: a newsletter's masthead and columns, a parish magazine's masters and styles. File › Save as template… saves the document as one, and File › New from template starts a new, untitled document from it. Two come with the program:

- **Newsletter**: a Letter page with a masthead, two linked columns and page numbers in the footer.
- **Booklet**: eight A5 facing pages with one story running through them all, and page numbers on the outside corners.

Templates saved in the templates folder (where Save as template starts) are listed on the menu; any `.avpt` file can be opened as a template with A template file….

Saving, exporting and printing



Saving. Ctrl+S saves, Ctrl+Shift+S saves under a new name. Saving writes the new file before it replaces the old one, so a failed save never damages your work. The title bar shows a dot while there are changes not saved, and closing asks first. File › Open recent lists the last ten documents. Undo (Ctrl+Z) and redo (Ctrl+Y) go all the way back to when the document was opened.

PDF. File › Export PDF… (Ctrl+E) writes the pages as a PDF, with its fonts inside it, ready to send to a printer or post online. Choose all the pages, the one shown, or a range such as `1-3, 7`.

PNG. File › Export PNG… writes each page as a picture, at a resolution from 72 to 600 dots an inch. Several pages go to numbered files: `leaflet-1.png`, `leaflet-2.png`.

Printing. File › Print… (Ctrl+P) sends the pages to your printer through the computer's own printing (on Linux and macOS, the `lp` command).

Booklets. Tick **As a booklet** when exporting a PDF or printing, and the pages are arranged two to a sheet in the order that folds into a stapled booklet. Page 8 goes beside page 1, page 2 beside page 7, and so on, with blank pages added to make a multiple of four. Print both sides of the paper, flipping on the short edge, fold the stack in half and staple it on the fold.

Keyboard

Keys	
V, T, R, E, L, P, B, G, K, H	The tools (when not typing)
Ctrl+N, Ctrl+O, Ctrl+S, Ctrl+Shift+S	New, open, save, save as
Ctrl+E, Ctrl+P	Export PDF, print
Ctrl+Z, Ctrl+Y	Undo, redo
Ctrl+X, Ctrl+C, Ctrl+V	Cut, copy, paste (objects, or text while typing)
Ctrl+D	Duplicate
Ctrl+G, Ctrl+Shift+G	Group, ungroup
Ctrl+], Ctrl+[	Bring forward, send backward (with Shift: to the front, to the back)
Arrows	Nudge by a point (with Shift: ten)
Ctrl+A	Select everything on the page (or the whole story while typing)
Ctrl+0, Ctrl+1, Ctrl+wheel	Fit the page, actual size, zoom
Page Up, Page Down	The previous and next page (while typing, the previous and next frame)
Ctrl+B, Ctrl+I, Ctrl+U	Bold, italic, underline
Alt+Left, Alt+Right	Kern tighter, looser
Ctrl+Shift+3	Insert a page number
Ctrl+F, F3, Shift+F3	Find and replace, find the next, the one before
F7	Check spelling
Ctrl+Shift+E	The story editor
Tab	In a table, the next cell
Space (held)	Slide the page about
Escape	Stop typing; then let go of the selection; or give up a drag

Help › Keys and tools has the short version.

Where things are kept

Your settings, your own dictionary (`words.txt`) and your templates are kept in the AVAPageStream folder of your user settings: `~/.config/AVAPageStream` on Linux and macOS, and `%APPDATA%\AVAPageStream` on Windows.